Challenge – Skin lesion classification

Teams of **two or three**. You train a model that recognises 7 kinds of skin lesions, make it usable in a small app, and present what you tried. For the leaderboard, I (Annika) will test every model on the official test images with my own script.

The data: DermaMNIST

DermaMNIST (MedMNIST v2, Yang et al., *Scientific Data* 2023) is based on the HAM10000 dermatoscopy images: **28 × 28 colour images**, **7 classes**. Load it with `pip install medmnist` – the starter notebook does it for you. Official splits: **7 007 train**, **1 003 val**, 2 005 test (licence CC BY-NC 4.0).

label	class	training images	share
0	actinic keratoses and intraepithelial carcinoma	228	3 %
1	basal cell carcinoma	359	5 %
2	benign keratosis-like lesions	769	11 %
3	dermatofibroma	80	1 %
4	melanoma	779	11 %
5	melanocytic nevi (harmless moles)	4 693	**67 %**
6	vascular lesions	99	1 %

Strongly imbalanced! A model that always says "nevus" has 67 % accuracy – but finds no melanoma. Therefore the **main metric is balanced accuracy** (= mean sensitivity over the 7 classes) on the official test set (the leaderboard). Also report **macro-F1** and the **sensitivity per class**.

What you do

A – Baseline. Run and understand the starter notebook `day4_challenge_starter.ipynb`: data, baseline CNN, training loop, evaluation, export. Describe the data (class counts, example images) and the baseline results.

B – Improve it. At least **four** well-motivated improvements, e.g. class weights, data augmentation, a bigger or better CNN (BatchNorm, dropout, more blocks), a learning-rate schedule, a pretrained ResNet. At least **two** must be motivated by a recent paper (e.g. NeurIPS 2023-2025) or another good source – cite it.

Compare fairly: change one thing at a time; train every variant with $\geq$ **3 seeds** and report mean $\pm$ std of the balanced accuracy on the **validation** images; give the **95 % bootstrap interval of the difference** to the baseline (the starter notebook has `bootstrap_difference`).

Track your experiments with **Weights & Biases**, **MLflow** or **ClearML** (free; recommended) or TensorBoard – a screenshot of your runs belongs in the report.

Make the **App** more creative / better for your model.

Rules

- **Train only on the train images**, choose models and settings on the **val** images.
- The official **test** split: you may use it to **report** the results of your *final* models (report, poster, presentation) – but **never train on it** and never use it to choose models or settings.
- Keep the input format of the starter: images `(N, 3, 28, 28)`, normalised with the given `MEAN`/`STD`.
- Any compute is fine (Colab, your own GPU, a cluster, …) – but **your code must run**: with your instructions and package list, a fresh run reproduces your model. **Important!**
- AI tools (ChatGPT, Copilot, …) are allowed – **if you say how you used them** in the report (AI use statement). Undisclosed use counts as an attempt to deceive.
- All team members must be able to explain every part of the code.

☑ What you hand in

- ☐ **`model.pt`** – exported with TorchScript (section 4 of the starter): input normalised `(N, 3, 28, 28)` float tensor, output logits `(N, 7)`. Check that it loads and runs (the starter does it).
- ☐ **Code** (Git repository) that reproduces your model from the data - including the list of dependencies.
- ☐ **App** from our template `app_template.py` (Gradio): upload an image → top-3 classes with probabilities. Runs in Colab or on Hugging Face Spaces; you show it **live** in the presentation.
- ☐ **Short report, 2–4 pages, IEEE conference format** (LaTeX template `report_template/`, also works in Overleaf): data, baseline, your experiments, results table (mean ± std, interval vs. baseline), discussion (what worked, what not, why), AI use statement, and at the end a short **contribution statement**: who did what (2–3 sentences).
- ☐ **Poster** (printed, portrait, size **A3, A2 or A1**): the problem, your model, the results table and one key figure – it hangs next to you during the presentation.
- ☐ **Presentation of the poster** with live demo, about **12 minutes** + questions, on the last day.

🗓 Timeline

when	what
23 Oct	kick-off: teams, starter notebook together, first baseline run
until 26 Oct	send me your team name and the team members (email: annika.stiehl@hs-ansbach.de)
4 + 5 Nov 2026	project days: work on your challenge in class (with help)
10 Nov 2026	send your `model.pt` to the lecturer · print your poster
11 Nov 2026	hand in code, app and report · poster presentations + leaderboard reveal

☆ Grading

part	weight
A: baseline and understanding of the data	10 %
B: improvements and their reasoning	25 %
evaluation: seeds, interval vs. baseline, right metrics	20 %
report	15 %
presentation, poster and questions	15 %
deployment (app)	5 %
code quality (runs, readable, reproducible)	10 %

The leaderboard rank gives at most a **small bonus** – a clean evaluation of a modest model is worth more than a lucky top score.